Question: What is the first and last useable host address on the network 172.26.160.0, subnet mask 255.255.255.192?

Binary IP Address: 10101100.00011010.10100000.00000000

Binary Subnet Mask: 11111111.11111111.11111111.11000000 (last 6 bits are host portion)

First Useable Address: 10101100.00011010.10100000.00000001

Last Useable Address: 10101100.00011010.10100000.00111110

Broadcast Address: 10101100.00011010.10100000.00111111

Second Useable Address: 10101100.00011010.10100000.00000010

First Useable Address:172.26.160.1

Last Useable Address:172.26.160.62

Useable host address range :172.26.160.1-172.26.160.62

Question: What is the last useable host address and Broadcast address for the ip address 172.27.16.64/22?

Binary IP Address: 1010110000.0110110.0010000.01000000

Binary Subnet Mask: 11111111.11111111.11111100.00000000

Network Address:

1010110000.0110110.0010000.00000000

Broadcast Address:

1010110000.0110110.0010011.11111111

Last Useable Host Address:

1010110000.0110110.0010011.11111110

Broadcast Address: 172.27.19.255

Last Useable Host: 172.27.19.254

Question: You decide to subnet 172.22.94.0/23 using /25 subnet mask.

How many subnets will this create?

How many useable host addresses are available on each subnet?

Number of subnets:

2^2=4

Number of Hosts

(2^7)-2=126

Question: Create multiple subnets from the 72.14.15.0/24 network. You need **5 subnets** with up to **30 hosts on each subnet**.

What subnet mask should you use?

List the Subnets created?

How many unused subnets are there?

subnet using /27 - $2^3$=8

($2^5$)-2= 30

Subnet Addresses (just showing the last octet):

.00000000(72.14.15.0/27)

.00100000(72.14.15.32/27)

.01000000(72.14.15.64/27)

.01100000(72.14.15.96/27)

.10000000(72.14.15.128/27)

.10100000(72.14.15.160/27)

.11000000(72.14.15.192/27)

.11100000(72.14.15.224/27)

Question: PC1 and PC2 are on the same LAN. You assign 192.168.187.195/26 to PC1 and 192.168.187.99/26 to PC2.

Pinging PC2 from PC1 does NOT work. Explain why?

192.168.187.195/26

Binary IP Address: 11000000.10101000.10111011.11000011

Network Address: 11000000.10101000.10111011.11000000

192.168.187.99/26

Binary IP Address: 11000000.10101000.10111011.01100011

Network Address: 11000000.10101000.10111011.01000000

DIFFERENT NETWORK ADDRESSES? ALL HOSTS ON A LAN MUST BE ON THE SAME NETWORK AND HAVE THE SAME NETWORK ADDRESS